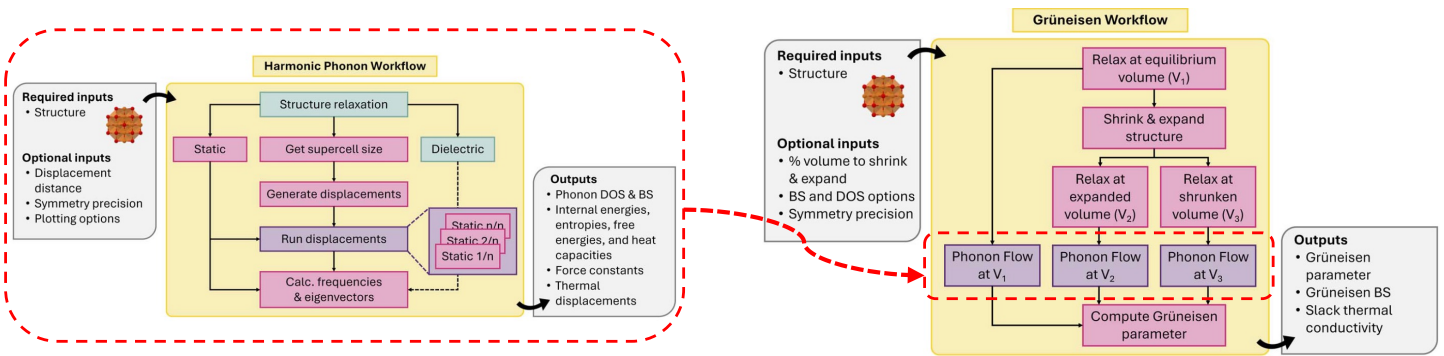

Harmonic Phonon Workflow
Required inputs
• Structure
Optional inputs
• Displacement distance
• Symmetry precision
• Plotting options
Structure relaxation
Static
Get supercell size
Dielectric
Generate displacements
Run displacements
Static n/n
Static 2/n
Static 1/n
Calc. frequencies & eigenvectors
Outputs
• Phonon DOS & BS
• Internal energies, entropies, free energies, and heat capacities
• Force constants
• Thermal displacements
Grüneisen Workflow
Required inputs
• Structure
Optional inputs
• % volume to shrink & expand
• BS and DOS options
• Symmetry precision
Relax at equilibrium volume (V_1)
Shrink & expand structure
Relax at expanded volume (V_2)
Relax at shrunken volume (V_3)
Phonon Flow at V_1
Phonon Flow at V_2
Phonon Flow at V_3
Compute Grüneisen parameter
Outputs
• Grüneisen parameter
• Grüneisen BS
• Slack thermal conductivity